AHMED AZIZ BARHOUMI

Manouba, Tunis | +216 58 714 686 | azizbr438@gmail.com | linkedin.com/in/aziz-barhoumi-320bb9339

Professional Profile

Third-year Engineering student in Applied Mathematics and Modelling at ENSIT, concurrently pursuing a Research Master's in Data Science. Hands-on project experience in Generative AI (LLM integration, RAG, agentic architectures with MCP and A2A) and end-to-end machine learning across NLP, speech, and computer vision. Seeking an AI / Generative AI / Data Science internship to apply strong mathematical foundations to practical, working systems.

Education

ENSIT, Tunis | *Research Master's Degree in Data Science* *2026 – 2027*

ENSIT, Tunis | *Engineering Degree in Applied Mathematics & Modelling* *2024 – 2027*

Preparatory Cycle (Physics-Chemistry) *2022 – 2024*

Baccalaureate in Experimental Sciences – Lycée Bab Elkhadhra, Tunis *2022*

Technical Skills

Generative AI & LLM Engineering: LLMs, RAG, Prompt Engineering, ChromaDB, LangChain, LangGraph, Model Context Protocol (MCP), Agent-to-Agent (A2A), n8n, Vision-Language Models, multi-provider LLM integration (Groq, Mistral, OpenAI, Whisper)

Data Science & Machine Learning: Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, TF-IDF, end-to-end ML pipelines

Deep Learning: PyTorch, TensorFlow, CNNs, CNN-LSTM, Transformers (BERT, NLP), Computer Vision, OCR

Programming Languages: Python, Java, C/C++, JavaScript, TypeScript, MATLAB, PL/SQL

Data & Databases: PostgreSQL, MySQL, MongoDB, web scraping (Selenium, BeautifulSoup)

Development & Tools: Docker, FastAPI, Flask, REST APIs, Streamlit, React, TailwindCSS, Git

Experience & Projects

AgriMent – Multi-Agent AI Platform for Farmers | *2nd Place, Talan* *2nd-Year Internship, 2 months*

- Led development of a generative AI decision-support platform in which specialized agents each cover one dimension of a farmer's decision:
  - **Agricultural Advisor Agent:** LLM-based crop and farming guidance.
  - **Regulatory Advisor Agent:** answers compliance questions from a regulatory knowledge base (PAC, TelePAC, FranceAgriMer, CERFA).
  - **Financial / Business Advisor Agent:** supports economic and business decisions.
  - **Pest Detection component:** VLM-based pest identification from images, feeding pest alerts.
- Built the FastAPI backend end to end, integrating farmer data, pest alerts and agent recommendations into a single working deployment.

NeuroLearn AI – Multi-Agent Adaptive STEM Learning Platform | *IEEE Education Week Challenge* *2026*

- Built a multi-agent generative AI learning platform that adapts STEM content to each learner through cognitive profiling (85%+ profile detection accuracy), with LLM routing across Groq, Mistral and OpenAI Whisper.
- Designed specialized learning agents, each turning the learner's cognitive profile into a different learning experience:
  - **Visual Agent:** generates visual explanations of STEM concepts for visual learners.
  - **Game Zone Agent:** turns course content into interactive, game-based exercises.
  - **Storytelling Agent:** explains concepts through generated narratives.
  - **Speech processing:** voice interaction via OpenAI Whisper.

City Brain Convergence – Agentic AI for Smart Cities *IEEE TSYP14 Technical Challenge*

- Engineered an agentic AI system using multi-agent architectures and Agent-to-Agent (A2A) protocols to continuously monitor digital-twin sensor streams and detect real-time urban incidents (accidents, fires, traffic, flooding).
- Autonomously triggered targeted physical and digital actuators for rapid emergency response.

FakeGuard AI – Fake News Detection System | *End-of-Second-Year Project* *2026*

- Built a full-pipeline fake news classifier on the WELFake dataset (tokenization, TF-IDF, transformer encoding); BERT outperformed CNN-LSTM and classical baselines.
- Developed a React web application accepting multi-format input (text, URLs, PDFs, images).

Emotion Recognition from Speech | *End-of-First-Year Project* *2025*

- Built an audio classification pipeline using Librosa (MFCCs, ZCR, spectral features); trained and compared SVM, Random Forest, XGBoost, and a hybrid CNN-LSTM architecture.

Data Extraction Tool – ProVerdy | *1st-Year Internship, 2 months* *2024*

- Developed an automated scraping tool with Selenium and BeautifulSoup to extract financial and sustainability reports from MENA sources and LinkedIn profiles, within platform policy limits.

Certifications

NVIDIA: Fundamentals of Deep Learning

365 Data Science: Introduction to Data Engineering; Introduction to AI

Leadership & Languages

Leadership: Human Resources Manager, IEEE ENSIT Student Branch

Languages: Arabic (Native), English (Intermediate), French (Intermediate)